Started on	Saturday, 10 October 2026, 6:21 PM
State	Finished
Completed on	Saturday, 10 October 2026, 6:27 PM
Time taken	5 mins 58 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main() {
    int n,k;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++) {
        scanf("%d",&a[i]);
    }
    scanf("%d",&k);
    if(k==0) {
        for(int i=1;i<n;i++) {
            if(a[i]==a[i-1]) {
                printf("1");
                return 0;
            }
        }
        printf("0");
        return 0;;
    }
    int i=0,j=1;
    while(i<n && j<n) {
        if(i==j) {
            j++;
        }
        else if(a[j]-a[i]==k) {
            printf("1");
            return 0;
        }
        else if(a[j]-a[i]<k)
        j++;
        else
        i++;
    }
    printf("0");
    return 0;

}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.